EM and the Forward–Backward Algorithm for Hidden Markov Models

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Methods in Computational Neuroscience, MBL — 2026

Abstract

These notes derive the forward-backward algorithm and EM (Baum-Welch) [Baum et al., 1970] for hidden Markov models directly, by maximum likelihood. This is the machinery underneath every `.fit(method="em")` call in this afternoon's coding tutorial — today we derive what that call is actually doing, down to the weighted linear regression at the heart of the tutorial's AR-HMM M-step. Based on `stats305c/em` and `stats305c/hmms`.

Contents

1 Hidden Markov Models

A **hidden Markov model (HMM)** pairs a Markov chain of discrete latent states $z_1, \dots, z_T \in \{1, \dots, K\}$ with a sequence of observations $x_1, \dots, x_T$, one emitted per state. The joint distribution factors as

$$p(z_{1:T}, x_{1:T}; \Theta) = p(z_1; \pi_0) \prod_{t=2}^{T} p(z_t \mid z_{t-1}; P) \prod_{t=1}^{T} p(x_t \mid \theta_{z_t}), \tag{1}$$

with three parameter groups $\Theta = (\pi_0, P, \{\theta_k\}_{k=1}^K)$:

1. **Initial distribution:** $z_1 \sim \mathrm{Cat}(\pi_0)$.
2. **Transition matrix:** $P \in [0,1]^{K \times K}$ (row-stochastic), $P_{ij} = p(z_{t+1}{=}j \mid z_t{=}i)$.
3. **Emissions:** $x_t \sim p(\cdot \mid \theta_{z_t})$ — e.g. Gaussian, Poisson, or, as in the MoSeq model [Wiltschko et al., 2015] and this afternoon's tutorial, an *autoregressive* Gaussian , $p(x_t \mid \theta_k) = \mathcal{N}(x_t \mid A_k x_{t-1} + b_k, Q_k)$, where $\theta_k = (A_k, b_k, Q_k)$.

We want the **maximum likelihood estimate** of Θ: the parameters that make the observed data $x_{1:T}$ most probable, marginalizing over the states we never observe,

$$\hat{\Theta} = \arg\max_{\Theta} \; \log p(x_{1:T}; \Theta), \qquad p(x_{1:T}; \Theta) = \sum_{z_{1:T}} p(z_{1:T}, x_{1:T}; \Theta). \tag{2}$$

The sum in (2) is over all K^T possible state sequences — already intractable to evaluate for, say, $K = 6$ and a few hundred time steps, let alone to differentiate for gradient ascent. **EM** sidesteps this by exploiting the Markov structure of (1): it turns out we only ever need two low-dimensional summaries of the posterior over paths, not the whole K^T-entry table, and those summaries can be computed by the **forward-backward algorithm** [Rabiner and Juang, 1986] in time that is *linear* in the length of the sequence, T. That's the plan for the rest of these notes: derive EM in general, including the M-step for a simple Gaussian emission model (Section 2), then derive forward-backward as the way to compute the summaries EM's E-step needs (Section 3). The M-step for the tutorial's autoregressive emissions is left as an exercise at the end of Section 2.

2 The EM Algorithm

2.1 The Evidence Lower Bound (ELBO)

Let's start by expanding the objective — the marginal log likelihood — and applying one of the oldest tricks in the book, multiplying by one. Introduce an **auxiliary distribution** $q(z_{1:T})$ over the whole hidden path — for now, completely arbitrary, with the same support as the true posterior. Multiply and divide by q inside the sum in (2):

$$\begin{aligned}
\log p(x_{1:T}; \Theta) &= \log \sum_{z_{1:T}} p(z_{1:T}, x_{1:T}; \Theta) \\
&= \log \sum_{z_{1:T}} q(z_{1:T}) \, \frac{p(z_{1:T}, x_{1:T}; \Theta)}{q(z_{1:T})} \\
&= \log \mathbb{E}_q \left[\frac{p(z_{1:T}, x_{1:T}; \Theta)}{q(z_{1:T})} \right]
\end{aligned} \tag{3}$$

Now that we've written the objective as the log of an expectation, we can apply **Jensen's inequality**.

Lemma (Jensen's inequality)**.** *For any concave function f and random variable Y, $f(\mathbb{E}[Y]) \geq \mathbb{E}[f(Y)]$, with equality iff Y is constant (or f is linear). Since* $\log$ *is concave,* $\log \mathbb{E}[g] \geq \mathbb{E}[\log g]$ *for any positive random variable g.*

Applying Jensen's inequality to (3), we obtain

$$\log p(x_{1:T}; \Theta) \geq \mathbb{E}_q \Big[\log p(z_{1:T}, x_{1:T}; \Theta) - \log q(z_{1:T}) \Big] \triangleq \mathcal{L}(\Theta, q). \tag{4}$$

$\mathcal{L}(\Theta, q)$ is the **evidence lower bound (ELBO)**. It holds for *any* q, so we've replaced the hard problem of maximizing $\log p(x_{1:T}; \Theta)$ with a potentially easier problem: maximize $\mathcal{L}(\Theta, q)$ over (Θ, q) by coordinate ascent. That coordinate ascent is EM — the E-step optimizes q and the M-step optimizes Θ — and it can be preferable to direct gradient ascent on (2) because, as we'll see, each coordinate update often has a simple, closed form.

2.2 The E-step

Note that $\log p(z_{1:T}, x_{1:T}; \Theta) = \log p(z_{1:T} \mid x_{1:T}; \Theta) + \log p(x_{1:T}; \Theta)$. Expanding the joint distribution inside the expectation in (4) and rearranging terms — a procedure we might call *ELBO surgery* — we obtain another form of the bound,

$$\mathcal{L}(\Theta, q) = \log p(x_{1:T}; \Theta) - D_{\mathrm{KL}}\big(q(z_{1:T}) \,\|\, p(z_{1:T} \mid x_{1:T}; \Theta)\big), \tag{5}$$

where $D_{\mathrm{KL}}(q\|p) = \sum_z q(z) \log \frac{q(z)}{p(z)}$ is the **Kullback-Leibler (KL) divergence**. Since the KL divergence is non-negative and zero iff $q = p$, we see that $\mathcal{L} \le \log p(x_{1:T}; \Theta)$ always, and maximizing $\mathcal{L}$ over q (with Θ fixed) means minimizing that KL divergence. The unique minimizer is

$$\boxed{q^\star(z_{1:T}) = p(z_{1:T} \mid x_{1:T}; \Theta),} \tag{6}$$

the exact posterior over the whole path, at which point the bound is tight: $\mathcal{L}(\Theta, q^\star) = \log p(x_{1:T}; \Theta)$. This is the **E-step**.

The catch: $q^\star$ is a distribution over K^T paths — we can't just write down a table of that size, let alone use it directly in the M-step below. Fortunately, as we will see, the M-step only needs *posterior marginals*, and we can compute those efficiently thanks to the Markovian nature of the model.

2.3 The M-step

The M-step maximizes the ELBO with respect to the parameters Θ, holding the posterior $q^\star$ fixed. Let's focus on the emission parameters $\{\theta_k\}_{k=1}^K$. As a function of those parameters, the ELBO is

$$\begin{aligned}
\mathcal{L}(\Theta, q) &= \mathbb{E}_q \left[\log p(x_{1:T}, z_{1:T}; \Theta)\right] + \mathrm{c} &(7)\\
&= \sum_{t=1}^T \sum_k \underbrace{q(z_t = k)}_{\triangleq \gamma_t(k)} \log p(x_t; \theta_k) + \mathrm{c}, &(8)
\end{aligned}$$

where $\gamma_t(k)$ is the probability of being in state k at time t under the posterior q. Maximizing with respect to the parameters θ_k is simply a weighted maximum likelihood estimation problem. Often, the solution can be found in closed form.

Example: Gaussian emissions Let's consider a simple case where the emission model is Gaussian with mean θ_k,

$$p(x_t; \theta_k) = \mathcal{N}(x_t \mid \theta_k, \sigma^2 I). \tag{9}$$

The ELBO as a function of θ_k is

$$\begin{aligned}
\mathcal{L}(\Theta, q) &= \sum_{t=1}^T \gamma_t(k) \log \mathcal{N}(x_t \mid \theta_k, \sigma^2 I) + \mathrm{c} &(10)\\
&= \sum_{t=1}^T \gamma_t(k) \left(-\frac{1}{2\sigma^2}(x_t - \theta_k)^\top (x_t - \theta_k)\right) + \mathrm{c}. &(11)
\end{aligned}$$

It's a little bit of calculus to show that this objective is maximized at

$$\hat{\theta}_k = \frac{\sum_{t=1}^T \gamma_t(k) x_t}{\sum_{t=1}^T \gamma_t(k)}. \tag{12}$$

Intuitively, we simply set the mean parameter equal to the average of observations, weighted by the probability of being in state k.

Exercise Now consider an autoregressive observation model (dropping the bias term for simplicity; see the remark below), $p(x_t \mid x_{t-1}; \theta_k) = \mathcal{N}(x_t \mid A_k x_{t-1}, Q_k)$. Show that the optimal dynamics parameters are obtained by the **weighted normal equations**, with solution

$$\boxed{\hat{A}_k = \left(\sum_{t=1}^{T} \gamma_t(k)\, x_t x_{t-1}^\top\right) \left(\sum_{t=1}^{T} \gamma_t(k)\, x_{t-1} x_{t-1}^\top\right)^{-1}.} \tag{13}$$

This is exactly **weighted least squares**: regress x_t on x_{t-1}, weighting time step t by how confident the E-step is that $z_t = k$. (The tutorial's actual fit includes a bias term too, $A_k x_{t-1} + b_k$; recover it with the usual trick for affine regression — augment x_{t-1} with a constant 1 and fold b_k into A_k as an extra column.)

2.4 EM as a minorize-maximize algorithm

Each iteration:

1. **E-step** (minorize) — build $\mathcal{L}(\Theta^{(i)}, q)$, which touches $\log p(x_{1:T}; \Theta)$ at $\Theta = \Theta^{(i)}$;
2. **M-step** (maximize) — set $\Theta^{(i+1)} = \arg\max_\Theta \mathcal{L}(\Theta, q^{(i)})$.

Since the M-step can only increase $\mathcal{L}$, and $\mathcal{L} \leq \log p(x_{1:T}; \Theta)$ always, the marginal log-likelihood is **guaranteed to increase monotonically** every iteration, with no step size to tune. EM converges to a *local* optimum, and (like K-means) is sensitive to initialization; random restarts are standard practice.

3 The Forward-Backward Algorithm

Our target quantity is the marginal $\gamma_t(k) = p(z_t{=}k \mid x_{1:T})$ needed for the M-step above. The key to computing it efficiently is the following factorization.

Claim:

$$p(z_t{=}k \mid x_{1:T}) \;\propto\; \underbrace{p(z_t{=}k \mid x_{1:t-1})}_{\triangleq\, \alpha_t(k)} \; \underbrace{p(x_{t:T} \mid z_t{=}k)}_{\triangleq\, \beta_t(k)}. \tag{14}$$

Proof, by one application of Bayes' rule. Split $x_{1:T} = (x_{1:t-1}, x_{t:T})$ and apply Bayes' rule, treating $x_{t:T}$ as newly-arrived evidence added to $x_{1:t-1}$:

$$p(z_t{=}k \mid x_{1:t-1}, x_{t:T}) \;\propto\; p(x_{t:T} \mid z_t{=}k, x_{1:t-1})\, p(z_t{=}k \mid x_{1:t-1}).$$

In an HMM, once we know z_t, everything that happens from t onward is independent of everything that happened before — the defining Markov property — so $p(x_{t:T} \mid z_t{=}k, x_{1:t-1}) = p(x_{t:T} \mid z_t{=}k) = \beta_t(k)$, and $p(z_t{=}k \mid x_{1:t-1}) = \alpha_t(k)$ by definition. Multiplying the two together gives (14). $\square$

So the whole problem reduces to computing two quantities recursively: the **forward message** $\alpha_t(k) \triangleq p(z_t{=}k \mid x_{1:t-1})$ and the **backward message** $\beta_t(k) \triangleq p(x_{t:T} \mid z_t{=}k)$.

3.1 The forward algorithm

Claim: Let $\alpha_1(k) = p(z_1 = k) = \pi_{0,k}$. Then, for $t = 1, \ldots, T-1$, the forward messages can be computed recursively as, $\alpha_{t+1}(j) = \dfrac{1}{C_t} \sum_{k=1}^{K} P_{kj}\, \alpha_t(k)\, p(x_t \mid z_t{=}k)$, where $C_t \triangleq \sum_k \alpha_t(k)\, p(x_t \mid z_t{=}k)$.

Proof, by induction on t. *Base case:* $\alpha_1(k) = p(z_1{=}k \mid x_{1:0}) = p(z_1{=}k) = \pi_{0,k}$ — there is no data to condition on before $t = 1$.

Inductive step: suppose $\alpha_t(k)$ is known for every k. First, incorporate the new observation x_t via Bayes' rule to get the *filtered* distribution,

$$p(z_t{=}k \mid x_{1:t}) = \frac{p(x_t \mid z_t{=}k)\, \alpha_t(k)}{\sum_{k'} p(x_t \mid z_t{=}k')\, \alpha_t(k')} = \frac{p(x_t \mid z_t{=}k)\, \alpha_t(k)}{C_t}.$$

Then predict one step ahead: by the law of total probability and the Markov property (z_{t+1} depends on the past only through z_t),

$$\alpha_{t+1}(j) = p(z_{t+1}{=}j \mid x_{1:t}) = \sum_k p(z_{t+1}{=}j \mid z_t{=}k)\, p(z_t{=}k \mid x_{1:t}) = \sum_k P_{kj}\, \frac{\alpha_t(k)\, p(x_t \mid z_t{=}k)}{C_t}. \qquad \square$$

Writing $\ell_t = [p(x_t \mid z_t{=}1), \ldots, p(x_t \mid z_t{=}K)]^\top$ for the vector of emission likelihoods at time t, this is the matrix-vector recursion

$$\alpha_{t+1} = \frac{P^\top(\alpha_t \odot \ell_t)}{C_t}, \qquad C_t = \alpha_t^\top \ell_t, \tag{15}$$

where $\odot$ denotes elementwise vector multiplication.

Exercise: Complexity Analysis Recall that the naive complexity of computing the marginal likelihood was $O(K^T)$. What is the computational complexity of the forward pass?

Normalization Notice that α_t is a conditional distribution of z_t, so the forward messages are always normalized to be non-negative and sum to one. This is helpful for numerical stability, and it's a point we'll return to in the backward pass.

Exercise: Marginal log likelihood Show that

$$C_t = p(x_t \mid x_{1:t-1}) \tag{16}$$

is exactly the one-step-ahead predictive likelihood, so by the chain rule we get the marginal log-likelihood for free:

$$\log p(x_{1:T}) = \sum_{t=1}^{T} \log C_t. \tag{17}$$

This is the quantity we track as "the log-likelihood" during EM.

3.2 The backward algorithm

Claim: Let $\beta_T(k) = p(x_T \mid z_T = k)$. Then the backward messages can be computed recursively as, $\beta_t(k) = p(x_t \mid z_t{=}k) \sum_{j=1}^{K} P_{kj}\, \beta_{t+1}(j)$.

Proof, by induction on t, downward from T. Base case: $\beta_T(k) = p(x_T \mid z_T{=}k)$ — just the likelihood of the final observation.

Inductive step: suppose $\beta_{t+1}(j)$ is known for every j. First, peel off x_t: since x_t depends only on z_t, it's independent of everything else once we condition on z_t,

$$\beta_t(k) = p(x_t, x_{t+1:T} \mid z_t{=}k) = p(x_t \mid z_t{=}k)\, p(x_{t+1:T} \mid z_t{=}k).$$

Then introduce z_{t+1} by marginalizing: by the law of total probability and the Markov property (the future depends on z_t only through z_{t+1}),

$$p(x_{t+1:T} \mid z_t{=}k) = \sum_{j=1}^{K} p(z_{t+1}{=}j \mid z_t{=}k)\, p(x_{t+1:T} \mid z_{t+1}{=}j) = \sum_{j=1}^{K} P_{kj}\, \beta_{t+1}(j).$$

Multiplying the two pieces back together,

$$\beta_t(k) = p(x_t \mid z_t{=}k) \sum_{j=1}^{K} P_{kj}\, \beta_{t+1}(j). \qquad \square$$

In matrix form, $\beta_t = \ell_t \odot (P\beta_{t+1})$ — note the order: matrix-multiply by P first, *then* weight by the current-step likelihood ℓ_t, the mirror image of the forward recursion's "weight, then matrix-multiply" order. The backward pass runs from $t = T$ down to $t = 1$.

A note on normalization. Unlike α_t, β_t is not a normalized distribution over k — it's a product of $T - t + 1$ likelihood terms, so it can just as easily underflow for large T. But we're free to rescale it: since $\gamma_t(k) \propto \alpha_t(k)\, \beta_t(k)$ only needs $\beta_t(k)$ up to an overall constant, one that doesn't depend on k, we can renormalize $\beta_t(\cdot)$ by any convenient constant B_t at every step of the recursion — e.g. $\beta_t(k) \leftarrow \beta_t(k)/B_t$ with $B_t = \sum_k \beta_t(k)$ — without changing $\gamma_t(k)$ once it's normalized over k. This is the same numerical trick as for α_t, just applied by hand here instead of falling out of the derivation automatically.

3.3 Putting it together

One forward pass and one backward pass give $\alpha_t(k)$ and $\beta_t(k)$ for every t, and hence, via (14), the posterior marginal

$$\gamma_t(k) = p(z_t{=}k \mid x_{1:T}) \;\propto\; \alpha_t(k)\, \beta_t(k). \tag{18}$$

This is the **E-step**, computed in $O(TK^2)$ time, without ever writing down the K^T-entry posterior over paths.

Exercise The transition-matrix M-step needs one more quantity: the pairwise marginal $\xi_t(i,j) \triangleq p(z_t{=}i, z_{t+1}{=}j \mid x_{1:T})$. Using the same conditional-independence argument as the proof of (14) — but now keeping *both* z_t and z_{t+1} fixed, rather than marginalizing the transition between them — show that

$$\xi_t(i,j) \;\propto\; \alpha_t(i)\, \ell_{t,i}\, P_{ij}\, \beta_{t+1}(j)$$

(normalize over i, j to get a probability). This is exactly the weight $\hat{P}_{ij} \propto \sum_{t=1}^{T-1} \xi_t(i,j)$ needs in the transition-matrix M-step.

References

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